Yanir Kleiman

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Experienced generative AI researcher with a strong background in computer graphics, animation, and VFX. I am currently a senior research scientist at Kyoso, previously at Meta AI (FAIR / GenAI).

I completed my PhD at Tel Aviv University, working on shape analysis, segmentation and shape matching. In recent years I have led research projects in generative 3D modeling, image and video generation, diffusion models, LLM / VLM training, and conversational agents.

I have prior in-depth practical experience in animation and VFX production.

Professional Experience

2025 - now **Senior Research Scientist, Kyoso**

Kyoso is a series A startup developing an AI creative agent for graphic design, motion graphics, and animation.

- Leading the animation and motion graphics research efforts.
- 0-to-1 development of an animation agent, including data collection, training, and evaluation.
- Combining agent skills with fine-tuned LLMs to produce animation plans, keyframe data, and video editing.
- Ownership of animation features, coordinating engineering and design teams to integrate research into user-facing product.

2020 - 2025 **Research Engineer, Meta AI**

Llama team:

- Llama 3 and Llama 4 multi-modal pre-training and post-training
- Procedural generation of training data for visual math problems
- Augmenting and improving training data with LLM/VLM as a judge
- Directly contributed to increase in model performance over visual reasoning benchmarks (MMMU, MathVista)

Selected projects at FAIR / GenAI:

- Meta 3D TextureGen (lead researcher, joint first author)
- Meta 3D AssetGen
- Replay dataset: aligned multi-modal multi-view videos for 3D reconstruction (joint first author)

Technical leadership of 3D shapes dataset management and curation:

- Evaluation and quality estimation of massive datasets from external vendors
- Ingestion and conversion of 2M shapes in various 3D formats
- Shape preprocessing and rendering setups
- PBR materials manipulation and verification
- Extracted millions of auxiliary assets (textures, materials, renders) for training purposes.

Contributions to Spark AR - a 3D studio software that artists use to create effects for mobile phones and headsets:

- Incorporated real time location tracking simulation in Spark Studio
- Integrated simulated 3D environments into Spark Studio

2019 - 2020 **Research Scientist, Shapes AI**

An early stage startup company that developed solutions in the space of 3D shapes and video-based visual reasoning. I led the research efforts in the 3D domain, which include generative modeling, point clouds processing, and 3D search.

- Developed a novel patch-based deep learning method to clean up noisy point clouds.
- Developed an end-to-end solution for generating thousands of textured shapes from a small set, which includes procedural modeling and texture generation.
- Delivered a set of 50K objects to a customer for robotics software training.
- Worked on 3D shapes search where the input is an image or a 3D shape.

2018 - 2019 **Software Developer, DNEG**
DNEG is a high-end visual effects company. I developed tools for the "on-set" department which manages and curates the reference footage on the set of a show.

- Developed and supported tools for browsing massive sets of 100K+ images which are used for every production in the company.
- Developed a deep learning method to detect color charts in images, automating a previously manual process.
- Developed a novel deep learning method to remove noise artifacts and increase quality of compressed source material (see 2019 publication).
- Worked with Maya, Nuke, Alembic, libraw, OpenEXR, and other common tools of the VFX industry, using mostly C++, Python, and MEL scripts.

2016 - 2017 **Post-doc Researcher, Laboratoire d'Informatique**, École Polytechnique, France.
Part of a computer graphics group led by Maks Ovsjanikov.

2011 - 2016 **PhD Student, Computer Science**, Tel Aviv University, Israel.
Computer graphics lab, under the supervision of Prof. Daniel Cohen-Or.

2008-2010 **3D Animation / VFX Artist, Various Projects**
Lighting, shading, rendering, visual effects, compositing, tracking, scripting.

- **Crew 972 (for Warner Bros.)**
  Looney Tunes Show, Roadrunner segments
- **Gravity Israel Visual Effects**
  Flagship broadcast commercials
- **"Deus" (sci-fi TV show)**
  Visual effects shots over live plates
- **Vancouver Film School**
  3D Animation and Visual Effects Diploma, graduated with honors

2008 **Algorithms Developer, MutualArt Inc.**
Developed and implemented automated text categorization and linking algorithms.

2005 - 2007 **Engineering Manager, Amobee Media Systems Ltd.**
Led a team of 5 engineers in an early stage ad-tech startup.

- People management, task planning, recruiting and training.
- .NET infrastructures (ASP.NET), Java, and SQL Server development.

Expertise

ML / AI Experience with diffusion models, LLMs, VLMs, and agentic workflows.

3D Geometry Working with meshes, point clouds, and implicit 3D representations (NeRF / SDF)

3D Software Deep knowledge of the content production pipeline, including advanced scripting in Blender, Maya, Nuke, and After Effects.

Academics

2016 **PhD in Computer Science**, Tel Aviv University, Israel.
Dissertation: Semantic Similarity and Correspondence of 3D Shapes and Images.

2005 **M.Sc. in Computer Science**, Tel Aviv Academic College, Israel
Graduated with honors.

2000 **B.Sc. in Math and Computer Science**, Tel Aviv University, Israel

Publications

*Joint first authors marked with **

2025 **InstanceGen: Image Generation with Instance-level Instructions**
Etai Sella, Yanir Kleiman, Hadar Averbuch-Elor
SIGGRAPH, 2025

2024 **The Llama 3 herd of models**
Abhimanyu Dubey et al. (500+ authors)
arXiv, 2024

2024 **Meta 3D Gen (technical report)**
*Raphael Bensadoun, *Tom Monnier, *Yanir Kleiman, Filippos Kokkinos, Yawar Siddiqui, Mahendra Kariya, Omri Harosh, Roman Shapovalov, Benjamin Graham, Emilien Garreau, Animesh Karnewar, Ang Cao, Idan Azuri, Iurii Makarov, Eric-Tuan Le, Antoine Toisoul, David Novotny, Oran Gafni, Natalia Neverova, Andrea Vedaldi
Meta Research, 2024

2024 **Meta 3D AssetGen: Text-to-mesh generation with high-quality geometry, texture, and pbr materials**
Yawar Siddiqui, Tom Monnier, Filippos Kokkinos, Mahendra Kariya, Yanir Kleiman, Emilien Garreau, Oran Gafni, Natalia Neverova, Andrea Vedaldi, Roman Shapovalov, David Novotny
NeurIPS, 2024

2024 **Meta 3D TextureGen: fast and consistent texture generation for 3d objects**
*Raphael Bensadoun, *Yanir Kleiman, Idan Azuri, Omri Harosh, Andrea Vedaldi, Natalia Neverova, Oran Gafni
arXiv, 2024

2023 **Replay: Multi-modal multi-view acted videos for casual holography**
*Roman Shapovalov, *Yanir Kleiman, *Ignacio Rocco, David Novotny, Andrea Vedaldi, Changan Chen, Filippos Kokkinos, Ben Graham, Natalia Neverova
ICCV, 2023

2019 **Boosting VFX Production with Deep Learning**
Yanir Kleiman, Simon Pabst, Patrick Nagle
ACM SIGGRAPH Talks, 2019

2018 **PCPNet: Learning Local Shape Properties from Raw Point Clouds**
*Paul Guerrero, *Yanir Kleiman, Maks Ovsjanikov, Niloy J. Mitra
Eurographics, 2018

2018 **Group optimization for multi-attribute visual embedding**
Qiong Zeng, Wenzheng Chen, Zhuo Han, Mingyi Shi, Yanir Kleiman, Daniel Cohen-Or, Baoquan Chen, Yangyan Li
Visual Informatics, 2018

2018 **Dance to the beat: Synchronizing motion to audio**
Rachele Bellini, Yanir Kleiman, Daniel Cohen-Or
Computational Visual Media, 2018

2018 **Robust Structure-Based Shape Correspondence**
Yanir Kleiman, Maks Ovsjanikov
Computer Graphics Forum, 2018

2017 **Region-Based Correspondence Between 3D Shapes via Spatially Smooth Biclustering**
Matteo Denitto, Simone Melzi, Manuele Bicego, Umberto Castellani, Alessandro Farinelli, Mario A. T. Figueiredo, Yanir Kleiman, Maks Ovsjanikov
ICCV 2017

2016 **Semantic Similarity and Correspondence of 3D Shapes and Images**
Yanir Kleiman
PhD Dissertation

2016 **Time-varying Weathering in Texture Space**
Rachele Bellini, Yanir Kleiman, Daniel Cohen-Or.
SIGGRAPH, 2016

2016 **Toward Semantic Image Similarity from Crowdsourced Clustering**
Yanir Kleiman, George Goldberg, Yael Amsterdamer, Daniel Cohen-Or.
CGI, 2016

2015	**SHED: Shape Edit Distance for Fine-grained Shape Similarity** Yanir Kleiman, Oliver van Kaick, Olga Sorkine-Hornung, Daniel Cohen-Or. SIGGRAPH Asia, 2015
2015	**DynamicMaps: Similarity-based Browsing through a Massive Set of Images** Yanir Kleiman, Dov Danon, Jasmin Felberbaum, Joel Lanir, Daniel Cohen-Or. CHI, 2015
2014	**Shape Segmentation by Approximate Convexity Analysis** Oliver van Kaick, Noa Fish, Yanir Kleiman, Shmuel Asafi, Daniel Cohen-Or. ACM Transactions on Graphics (TOG), 2014
2013	**Dynamic Maps for Exploring and Browsing Shapes** Yanir Kleiman, Noa Fish, Joel Lanir, Daniel Cohen-Or. SGP, 2013
2011	**Unsupervised co-segmentation of a set of shapes via descriptor-space spectral clustering** Oana Sidi, Oliver van Kaick, Yanir Kleiman, Hao Zhang, Daniel Cohen-Or. SIGGRAPH Asia, 2011
2007	**Paging with connections: FIFO strikes again** Leah Epstein, Yanir Kleiman, Jiri Sgall, Rob van Stee. Theoretical computer science, 2007

Awards and Fellowships

2016-2017	**Chateaubriand Fellowship for Postdoctoral Research**
2014-2015	**Google Focused Research Award** I was partly funded by this grant during my PhD studies.
2010	**Animex Visual Effects Award** Runner up for Best Visual Effects in Animex 2010 Festival.
2005	**Excellence Scholarship - Tel Aviv Academic College** Awarded for excellence during my M.Sc. studies.